$P(X < 95) = P(Z < -1.667)$

$= P(Z > 1.667)$

$= 0.5 - P(0 < Z < 1.67)$

$= 0.5 - 0.4525$

$= 0.0475$

No. of bulbs expected to expire in less than 95 days out of 900 bulbs is $900 \times .0475 = 43$ bulbs

Example 7.27

Assume that the mean height of soldiers is 69.25 inches with a variance of 9.8 inches. How many soldiers in a regiment of 6,000 would you expect to be over 6 feet tall?

Solution :

Let X be the height of soldiers follows normal distribution with mean 69.25 inches and standard deviation 3.13. then the soldiers over 6 feet tall (6ft × 12= 72 inches)

The standard normal variate

$$Z = \frac{X - \mu}{\sigma} = \frac{72 - 69.25}{3.13} = 0.8786$$

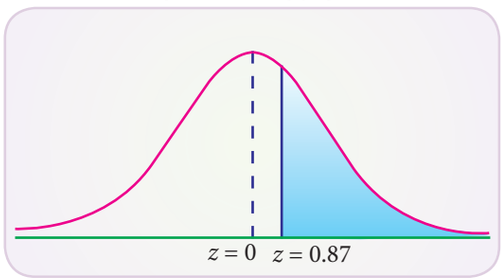


Fig 7.16

$P(X > 72) = P(Z > 0.8786)$

$= 0.5 - P(0 < Z < 0.88)$

$= 0.5 - 0.3106 = 0.1894$

Number of soldiers expected to be over 6 feet tall in 6000 are

$6000 \times 0.1894 = 1136$

Example 7.28

A bank manager has observed that the length of time the customers have to wait for being attended by the teller is normally distributed with mean time of 5 minutes and standard deviation of 0.7 minutes. Find the probability that a customer has to wait

(i) for less than 6 minutes

(ii) between 3.5 and 6.5 minutes

Solution :

Let X be the waiting time of a customer in the queue and it is normally distributed with mean 5 and SD 0.7.

(i) for less than 6 minutes

$$Z = \frac{X - \mu}{\sigma} = \frac{6 - 5}{0.7} = 1.4285$$

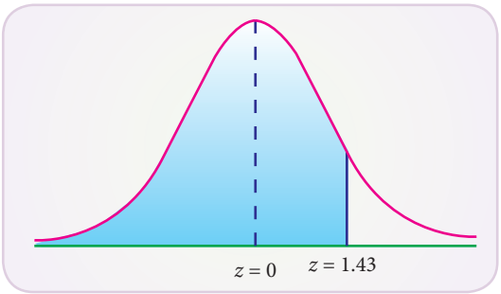


Fig 7.17

$P(X < 6) = P(Z < 1.43)$

$= 0.5 + 0.4236$

$= 0.9236$

(ii) between 3.5 and 6.5 minutes

When $X = 3.5$

$$Z = \frac{X - \mu}{\sigma} = \frac{3.5 - 5}{0.7} = -2.1429$$

When $X = 6.5$

$$Z = \frac{X - \mu}{\sigma} = \frac{6.5 - 5}{0.7} = 2.1429$$

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